

Legacy SQL Server Modernization Platform

Executive Summary

The Legacy SQL Server Modernization Platform enables organizations to migrate applications from Microsoft SQL Server to PostgreSQL with minimal application changes.

Rather than requiring costly rewrites of legacy applications, the platform provides:

1. Automated migration assessment
2. AI-assisted schema conversion
3. Intelligent SQL translation middleware
4. Long-term PostgreSQL compatibility support

The goal is to reduce SQL Server licensing costs, modernize infrastructure, and accelerate cloud migrations while minimizing business risk.

Problem Statement

Many organizations rely on legacy applications that were developed specifically for Microsoft SQL Server.

Common migration challenges include:

- Thousands of SQL queries embedded throughout applications
- Extensive use of stored procedures
- SQL Server-specific syntax and functions
- Limited documentation
- High costs associated with application rewrites
- Expensive SQL Server licensing and infrastructure

Existing solutions often require:

- Significant code changes
- Manual conversion efforts
- Complete application redevelopment
- Acceptance of compatibility limitations

Organizations need a lower-risk path to modernization.

Solution Overview

The platform provides a structured migration process consisting of four major phases.

Phase 1: Migration Assessment

Objective

Determine migration complexity before any changes are made.

Assessment Activities

Analyze:

- Database schema
- Stored procedures
- Views
- Triggers
- Indexes
- User-defined functions
- SQL Server-specific features
- Application query patterns
- ODBC and driver usage

Deliverables

Migration report including:

Metric	Description
Compatibility Score	Estimated migration readiness
Automated Conversion Percentage	Portion of system expected to migrate automatically
Manual Effort Estimate	Expected engineering effort
Risk Areas	SQL features requiring special handling
Cost Savings Estimate	SQL Server licensing and infrastructure savings

Example Output

Migration Readiness Score: 87%

Automatically Convertible: 91%

Requires Manual Review: 7%
Unsupported Features: 2%

Phase 2: AI-Assisted Schema Conversion

Objective

Convert Microsoft SQL Server schemas into optimized PostgreSQL schemas.

Conversion Scope

- Tables
- Indexes
- Constraints
- Relationships
- Data types
- Views
- Functions
- Stored procedures

AI Enhancements

Rather than relying on simple rules, AI evaluates:

- Naming conventions
- Business relationships
- Query history
- Existing usage patterns

Example

SQL Server:

```
CustomerID  
LastName  
OrderDate
```

PostgreSQL:

```
customer_id
last_name
order_date
```

The platform creates a complete translation dictionary that maps legacy structures to new PostgreSQL structures.

Phase 3: Intelligent Compatibility Middleware

Objective

Allow existing applications to continue functioning against PostgreSQL with minimal or no code changes.

Architecture

```
Legacy Application
  |
  ODBC
  |
Compatibility Middleware
  |
Translation Engine
  |
PostgreSQL
```

Core Features

T-SQL Translation

Converts SQL Server syntax into PostgreSQL-compatible syntax.

Examples:

SQL Server	PostgreSQL
TOP 10	LIMIT 10
ISNULL()	COALESCE()
GETDATE()	NOW()
LEN()	LENGTH()

Schema Translation Dictionary

Generated during migration.

Example:

```
{  
  "Customer": "customer_master",  
  "Order": "sales_order"  
}
```

This allows the middleware to perform deterministic translations instead of generic SQL parsing.

Query Rewriting

Example:

Legacy Query:

```
SELECT *  
FROM Customer
```

Translated Query:

```
SELECT *  
FROM customer_master
```

Metadata Mapping

Supports translation of:

- Table names
 - Column names
 - Views
 - Functions
 - Stored procedures
-

Phase 4: PostgreSQL Pass-Through Mode

Objective

Enable modernization without creating long-term vendor lock-in.

Benefits

New development can use native PostgreSQL syntax immediately.

Example:

Legacy Query:

```
SELECT TOP 10 *  
FROM Customer
```

Translated by middleware.

New Query:

```
SELECT *  
FROM customer_master  
LIMIT 10
```

Passed directly to PostgreSQL without modification.

Over time organizations can reduce dependency on the translation layer.

Query Learning Engine

Objective

Improve migration accuracy using real-world database usage.

Data Collection

Capture:

- SQL statements
- Stored procedure executions

- Query frequency
- Parameter usage
- Execution plans
- Object references

Benefits

Focus migration effort on:

- Frequently used tables
- Critical business processes
- High-value queries

Instead of attempting to migrate every object equally.

Competitive Differentiation

Traditional Migration Tools

Traditional tools typically:

- Convert schemas
- Generate scripts
- Leave application compatibility problems unresolved

Babelfish

Babelfish attempts generic SQL Server compatibility for all applications.

Challenges include:

- Metadata compatibility
- Stored procedure behavior
- SQL Server-specific features
- Driver expectations

Our Approach

The platform gains advantages through:

Application-Specific Knowledge

Every migration creates:

- Schema maps

- Query maps
- Object maps
- Translation rules

AI-Assisted Translation

Custom translation logic generated for each customer.

Risk Assessment First

Customers understand migration feasibility before committing.

Long-Term Modernization Path

Applications gradually move toward native PostgreSQL usage.

Target Customers

Primary Markets

Mid-Market Enterprises

Organizations seeking to reduce SQL Server licensing costs.

SaaS Vendors

Companies looking to standardize on PostgreSQL.

Cloud Migration Projects

Organizations migrating to AWS, Azure, or Google Cloud.

Private Equity Portfolio Companies

Businesses seeking infrastructure cost optimization after acquisition.

Revenue Model

Assessment Services

Fixed-fee migration assessments.

Example

- Small Database: \$5,000
 - Medium Database: \$15,000
 - Enterprise Database: \$50,000+
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Migration Services

Professional services engagement.

Example

- Fixed bid migration projects
 - Time and materials consulting
 - Managed migration engagements
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Software Licensing

Annual subscription for middleware.

Pricing Examples

- Small Business: \$10,000/year
 - Mid-Market: \$50,000/year
 - Enterprise: \$100,000+/year
-

Managed Service Offering

Hosted migration middleware and support.

Monthly recurring revenue model.

Product Roadmap

Phase 1

Assessment Platform

Features:

- Schema analysis
 - Risk scoring
 - Migration reports
-

Phase 2

AI Conversion Engine

Features:

- Schema conversion
 - Object mapping
 - Translation dictionary generation
-

Phase 3

Middleware MVP

Features:

- SELECT support
 - INSERT support
 - UPDATE support
 - DELETE support
-

Phase 4

Advanced Compatibility

Features:

- Stored procedure support
 - Metadata translation
 - Query optimization
 - Performance monitoring
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Vision

Enable organizations to migrate from SQL Server to PostgreSQL without expensive application rewrites.

The platform becomes the safest and fastest path for modernizing legacy database applications while reducing infrastructure costs and preserving business continuity.